

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 09

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?

A $y = -x^2 - 2x - 3$

B $y = -x^2 + 2x + 1$

C $y = x^2 - 4x + 5$

D $y = x^2 + 6x + 10$

E $y = x^2 + 4x + 1$

2 Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.

- A $\frac{13}{16}$
- B $\frac{16}{3}$
- C 1
- D $-\frac{1}{4}$
- E $-\frac{16}{3}$
- F $-\frac{1}{16}$

3 Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.

- A** π
- B** $\frac{5\pi}{6}$
- C** $\frac{7\pi}{6}$
- D** $\frac{\pi}{6}$
- E** $\frac{2}{3}\pi$
- F** $\frac{2\pi}{3}$

- 4 A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at
- A $(1, 3)$
 - B $(-3, 1)$
 - C $(-1, 3)$
 - D $(1, -3)$
 - E $(3, -1)$
 - F $(-1, -3)$

- 5 The curve $y = x^2 - 6x + 5$ is reflected in the y-axis. Find the equation of the reflected curve.
- A** $y = x^2 + 6x + 5$
- B** $y = x^2 + 6x - 5$
- C** $y = -x^2 - 6x + 5$
- D** $y = -x^2 + 6x - 5$
- E** $y = x^2 - 6x - 5$

- 6** £600 is divided between A , B and C in the ratio $2 : 3 : 7$. How much more does C receive than A ?
- A** £300
- B** £250
- C** £150
- D** £200
- E** £350
- F** £100

7 Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

A 16

B 6

C 9

D 18

E 24

F 8

8 Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

- A** $-\frac{32}{3}$
- B** $\frac{8}{3}$
- C** 8
- D** $\frac{16}{3}$
- E** $\frac{64}{3}$
- F** $\frac{32}{3}$

9 The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .

A 4

B 3

C $\sqrt{12}$

D 9

E 2

F 6

- 10** Given that $\int_1^5 f(x) \, dx = 7$ and $\int_3^5 f(x) \, dx = 2$, find $\int_1^3 (2f(x) - 1) \, dx$.
- A** 12
- B** 16
- C** 8
- D** 9
- E** 10
- F** 3

11 The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.

A 43

B 189

C 21

D 230

E 39

F 41

12 Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

- A** $\frac{2\pi}{3}$
- B** $\frac{4\pi}{3}$
- C** π
- D** $\frac{5\pi}{3}$
- E** 2π
- F** $\frac{7\pi}{3}$

13 The curve $y = ax^2 + bx$ has a stationary point at $(2, -8)$. Find $a + b$.

A -6

B -2

C 10

D 2

E -10

F 6

- 14** A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.

- A** $\frac{25}{216}$
B $\frac{5}{36}$
C $\frac{1}{216}$
D $\frac{6}{125}$
E $\frac{216}{91}$
F $\frac{216}{216}$

15 The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .

A 3

B -6

C ± 6

D 9

E 6

F 0

- 16** The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .
- A** $-2 < m < 3$
- B** $m > 3$
- C** $-6 < m < -2$
- D** $m < -2$ or $m > 3$
- E** $m > 0$
- F** $m > -6$

17 For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?

A 1

B 0

C 2

D 3

E infinitely many

- 18** The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4 - x}$. Which sequence of transformations achieves this?
- A** translate 4 units in the positive x -direction, then reflect in the y -axis
 - B** translate 4 units in the negative x -direction, then reflect in the y -axis
 - C** reflect in the x -axis, then translate 4 units in the positive y -direction
 - D** stretch parallel to the x -axis with factor 4
 - E** reflect in the y -axis, then translate 4 units in the positive x -direction

- 19** p is directly proportional to $\sqrt{q}$, and q is inversely proportional to r . Which of the following is true?
- A** p is directly proportional to $\sqrt{r}$
 - B** p is inversely proportional to r
 - C** p is inversely proportional to r^2
 - D** p is inversely proportional to $\sqrt{r}$
 - E** p is directly proportional to r

20 For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?

A 3

B -1

C $\frac{3}{2}$

D 6

E -3

F 1

END OF TEST